Microsoft Stock Data Analysis

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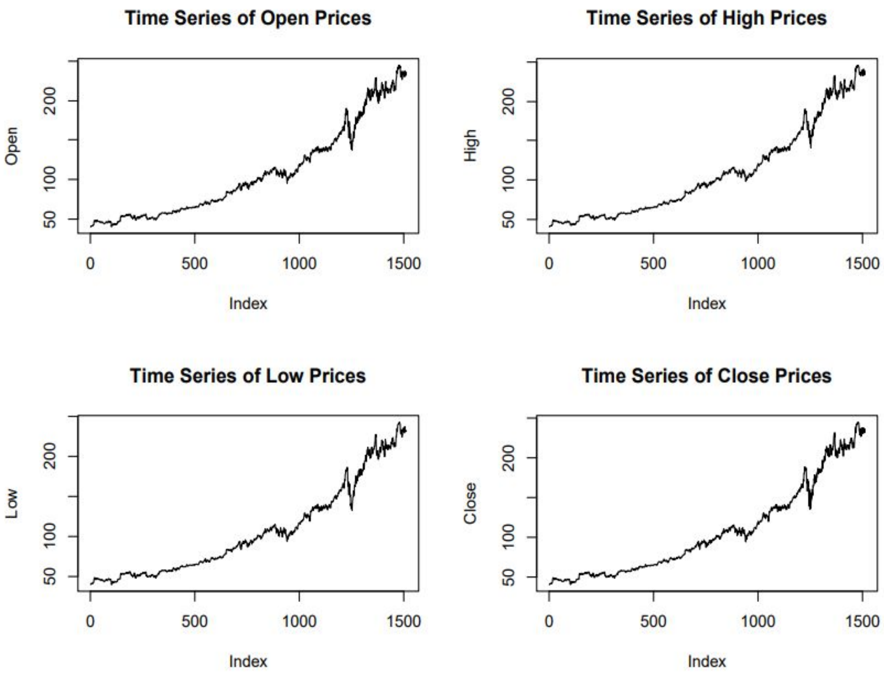
Introduction

- This report analyzes Microsoft’s stock data, covering variables like Open, High, Low, Close and Volume.
- Summary statistics : Mean, Median, Standard deviation.
- Key objectives :
- - Identify patterns, trends and randomness in stock prices
- - Use exploratory data analysis and statistical tests
- (Analysis of stock data, including exploratory data analysis, statistical tests, and non-parametric kernel regression.)

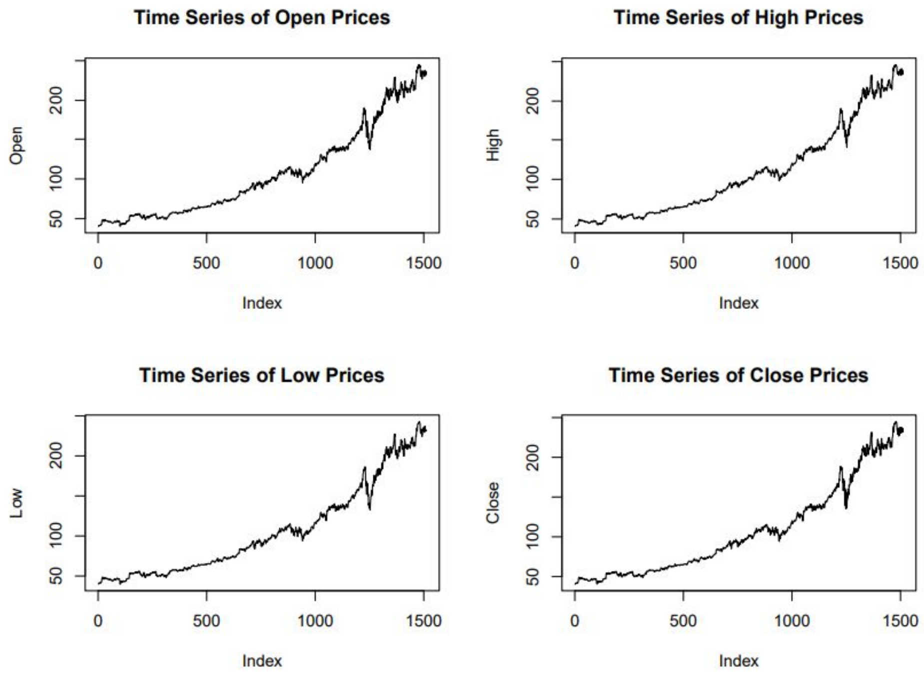
Exploratory Data Analysis

- Time Series plots of Microsoft's stock prices (Open, High, Low, Close) and Volume reveal significant trends.
- Observed upward trend across most variables.



Summary Statistics and Time Series Plots

```
      Open              High              Low
 Min.   : 40.34   Min.   : 40.74   Min.   : 39.72
 1st Qu.: 57.86   1st Qu.: 58.06   1st Qu.: 57.42
 Median : 93.99   Median : 95.10   Median : 92.92
 Mean   :107.39   Mean   :108.44   Mean   :106.29
 3rd Qu.:139.44   3rd Qu.:140.32   3rd Qu.:137.82
 Max.   :245.03   Max.   :246.13   Max.   :242.92
     Close            Volume
 Min.   : 40.29   Min.   :   101612
 1st Qu.: 57.85   1st Qu.: 21362129
 Median : 93.86   Median : 26629615
 Mean   :107.42   Mean   : 30198625
 3rd Qu.:138.97   3rd Qu.: 34319616
 Max.   :244.99   Max.   :135227059
```



Run Test for Randomness

- The run test assesses randomness in stock prices :
- Result : Non-random clustering over time in prices and volume
- Volume shows higher values in earlier periods, indicating inverse relationship with price trends.

Mann-Kendall Trend Test

- The Mann-Kendall test detects monotonic trends in the time series data :
- - Significant upward trends in price variables (Open, High, Low, Close)
- - Suggests a consistent rise in stock prices over time.

Run Test and Mann-Kendall Trend test

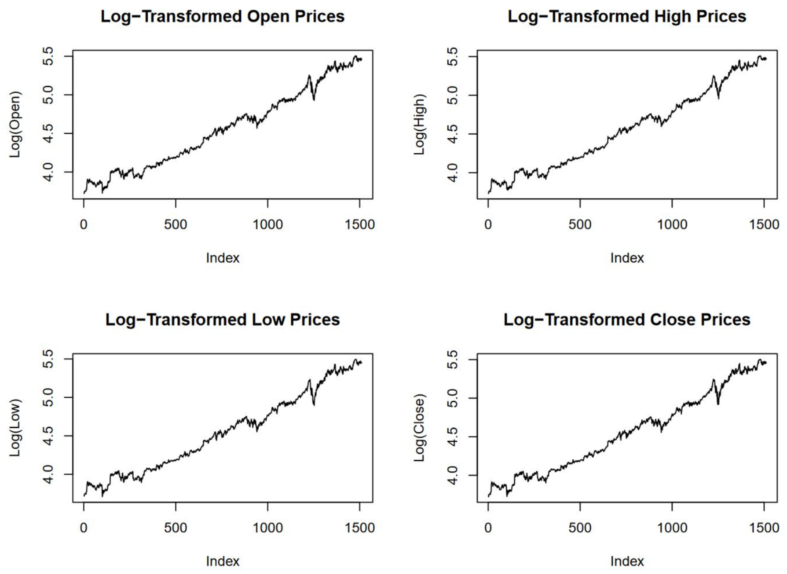
- Both Run test and Mann-Kendall trend test suggests significant monotonic upward Trend.
- To further analyze the data, we detrend each log-transformed variable by subtracting a 20-period moving average. This allows us to remove any remaining trend in the data.

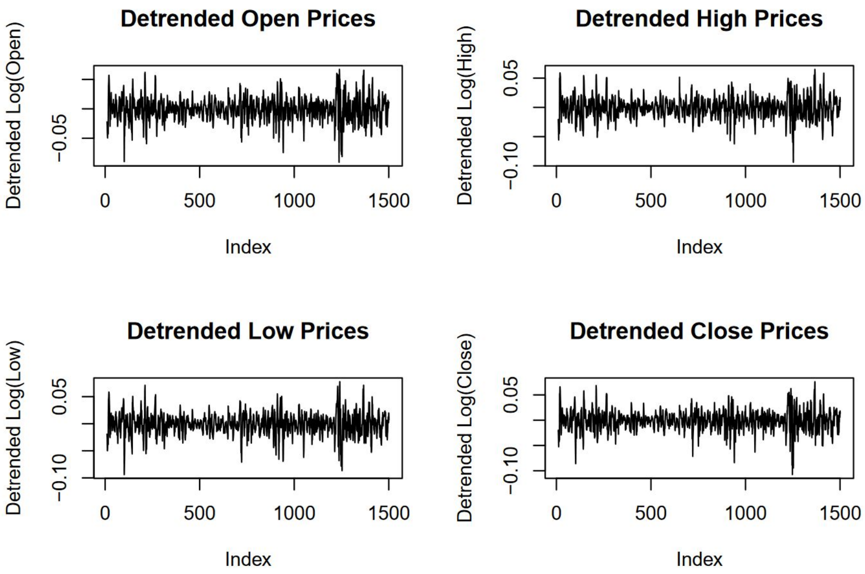
Log Transformation and Detrended Analysis

To reduce the exponential trends:

- Log Transformation applied to variables.
- Detrending with a moving average technique.
- Increased randomness post-detrend, allowing for further statistical tests.

Log Transformation and Detrended Analysis





Run Test on Detrended Data

- After detrending the data, higher p-value in Run test Suggests that the data now behaves more randomly.
- We still get low run but data seems to be more random. After removing the trend the run value is increased.
- We can now use non-parametric tests for our analysis, since sequence is random.

```
## Detrended Close Run Test on Detrended Data
## Test Statistic: -19.2161
## p-value: 2.714362e-82

## Detrended Open Run Test on Detrended Data
## Test Statistic: -20.14841
## p-value: 2.778728e-90

## Detrended High Run Test on Detrended Data
## Test Statistic: -21.28791
## p-value: 1.469257e-100

## Detrended Low Run Test on Detrended Data
## Test Statistic: -22.94537
## p-value: 1.639108e-116

## Detrended Volume Run Test on Detrended Data
## Test Statistic: -10.56626
## p-value: 4.271766e-26
```

Two-Sample Tests for Location and Scale

- Location and Scale comparisons were conducted:
- Scale Alternative - Ansari-Bradley Test
- Location Alternative - Wilcoxon Rank Sum Test
- Higher p-values indicate similar central tendencies and variability among variables.

Scale Alternative Test (Ansari-Bradley Test)

- Null Hypothesis: Scale parameter(theta) = 1
- **Close vs. Open**: The p-value (0.94) suggests that the variability between Close and Open prices is similar.
- **Close vs. High and Low**: Close prices show similar variability compared to both High ($p = 0.92$) and Low ($p = 0.94$).
- **Open vs. High and Low**: Open prices also have similar variability to both High ($p = 0.97$) and Low ($p = 0.87$).
- **High vs. Low**: The variability between High and Low prices is not significantly different ($p = 0.86$), indicating similar scale.

Location Alternative Test (Wilcoxon Rank Sum Test)

- Null Hypothesis: The two independent samples come from identical populations with equal medians.
- **Close vs. Open**: With a p-value of 0.99, the test shows no significant difference in the median between Close and Open prices.
- **Close vs. High and Low**: Close prices have similar medians to both High ($p = 0.52$) and Low ($p = 0.47$) prices.
- **Open vs. High and Low**: Open prices also show no significant difference in median compared to both High ($p = 0.51$) and Low ($p = 0.48$).
- **High vs. Low**: The median for High and Low prices is similar, as indicated by a p-value of 0.20.

Two-Sample Run Test for Distribution Comparison

- Null Hypothesis: feature_1 distribution = feature_2 distribution
- **Close vs. Open**: The test shows Close and Open prices have very similar distributions after removing trends ($p = 1.00$).
- **Close vs. High and Low**: Close prices are also very similar to both High ($p = 0.93$) and Low ($p = 0.89$) prices in distribution.
- **Open vs. High and Low**: Open prices are similar in distribution to both High ($p = 0.86$) and Low ($p = 0.84$) prices.
- **High vs. Low**: High and Low prices show the most difference, but still aren't significantly different in distribution ($p = 0.40$).

Spearman Rank Correlation

- Null Hypothesis: Feature 1 and Feature 2 are not Independent.
- Spearman rank correlation measures monotonic relationships :
- - High correlations between price variables (Open, High, Low, Close)
- - Low correlations between Volume and price variables, suggesting weaker relationships

Results from Spearman Rank Correlation test

Variables	Correlation_Coefficient	p_value
Close & Open	0.999404	0.0000000
Close & High	0.999678	0.0000000
Close & Low	0.999752	0.0000000
Close & Volume	0.006615	0.7972423
Open & High	0.999752	0.0000000
Open & Low	0.999647	0.0000000
Open & Volume	0.010142	0.6936468
High & Low	0.999621	0.0000000
High & Volume	0.014822	0.5648038
Low & Volume	0.001158	0.9641175

Complimentary Analysis on Cats and Dogs Dataset

- Additional non-parametric analysis on audio features from cats and dogs :
- - Compared features like Pitch, Spectral Centroid, and Zero Crossing Rate
- - Results show similar distribution characteristics between both datasets

Conclusion & Future Work

- Key insights and potential applications :
- - Run test recommended for model evaluation in machine learning (can be used as stopping criteria)
- - Extension of non- parametric methods to other complex data types (images, text)
- - Non-parametric tests offer flexibility for varied data distributions and fields